SafeStay Reports & Reviews

A short guide: what it is, how to run it, how to add it to another project

What this project is

SafeStay is a student accommodation system. This project adds the two admin pages that were missing, and connects them to a real MySQL database:

Page	What you can do
Reports	See reports made about accommodation (fake listing, scam, unsafe...). Filter by status, open a report, change its status (Pending, Under Review, Resolved, Rejected), add a new report, delete one, or export everything to a CSV file.
Reviews	See student reviews of accommodation with average ratings for safety, cleanliness, WiFi and landlord behaviour. Filter by accommodation, add, edit or delete a review.

What you need first

Java (version 11 or newer), Maven, and MySQL with MySQL Workbench. To check Java and Maven, type `java -version` and `mvn -version` in a terminal.

How to run it

1. **Create the database.** Open MySQL Workbench and connect to your server. Choose File, Open SQL Script, pick `SafeStay_Database.sql` and click the lightning-bolt button to run it.
2. **Add the sample data.** Do the same with `SafeStay_SeedData.sql`. This fills the pages with example reports and reviews.
3. **Check it worked.** In Workbench, run `SHOW TABLES;` in the `safestay` database. You should see 10 tables. If it is empty, step 1 did not finish; run it again.
4. **Enter your MySQL login.** Open `db.properties` in the project folder and type your MySQL password after `db.password=`. The user is usually `root`.
5. **Start the app.** Open a terminal in the project folder (the one that contains `pom.xml`) and run the command below. The first run downloads the MySQL driver, so it needs internet.

```
mvn compile exec:java
```

A window opens with two tabs: **Reports** and **Reviews**. You can also build one file with `mvn package` and start it with `java -jar target/safestay-app.jar`.

Using the pages

To do this	Do this
Open a report	Click View Report (or double-click the row). Change the status and press Save Status.
Add a report	Press New Report, choose the student and accommodation, give a reason, Submit.
Export reports	Press Export Reports and choose where to save the CSV file.

To do this	Do this
Add a review	On the Reviews tab press Add Review. Ratings are 1 to 5. A student can review each accommodation only once.
Edit or delete a review	Select a row, then Edit Selected or Delete Selected (or double-click a row to edit).

Adding it to another Java project

The code is split so it can be dropped into someone else's project. The screens never talk to the database directly; they go through a service layer, which uses the database layer.

1. Copy the folder `src/main/java/com/safestay` into the other project's source folder, keeping the folder names.
2. Add the MySQL driver to that project (in Maven: dependency `com.mysql:mysql-connector-j:8.4.0`; otherwise add the jar to the libraries).
3. Run the two SQL files on the same MySQL server.
4. Add these lines where the other project builds its window:

```
SafeStayModule module = new SafeStayModule(new DatabaseConnection());
tabs.addTab("Reports", module.createReportsPanel());
tabs.addTab("Reviews", module.createReviewsPanel());
```

Both pages are normal Swing panels, so they fit in a tabbed pane or any other container. If the other project uses a different package name, rename the packages with your editor's Refactor, Rename option so every import updates. Full details are in README.md.

If something goes wrong

Problem	Fix
Error saying a table does not exist	The database is empty. Run SafeStay_Database.sql, then SafeStay_SeedData.sql.
"Database Connection Failed" window	MySQL is not running, or the password in db.properties is wrong.
Same error with the right password	Start the app from the project folder; that is where db.properties is read from.
Want to start again	In Workbench run DROP DATABASE SAFESTAY; then redo steps 1 and 2.

Keep your real password out of anything you share. The zip supplied with this guide has the password field left empty.